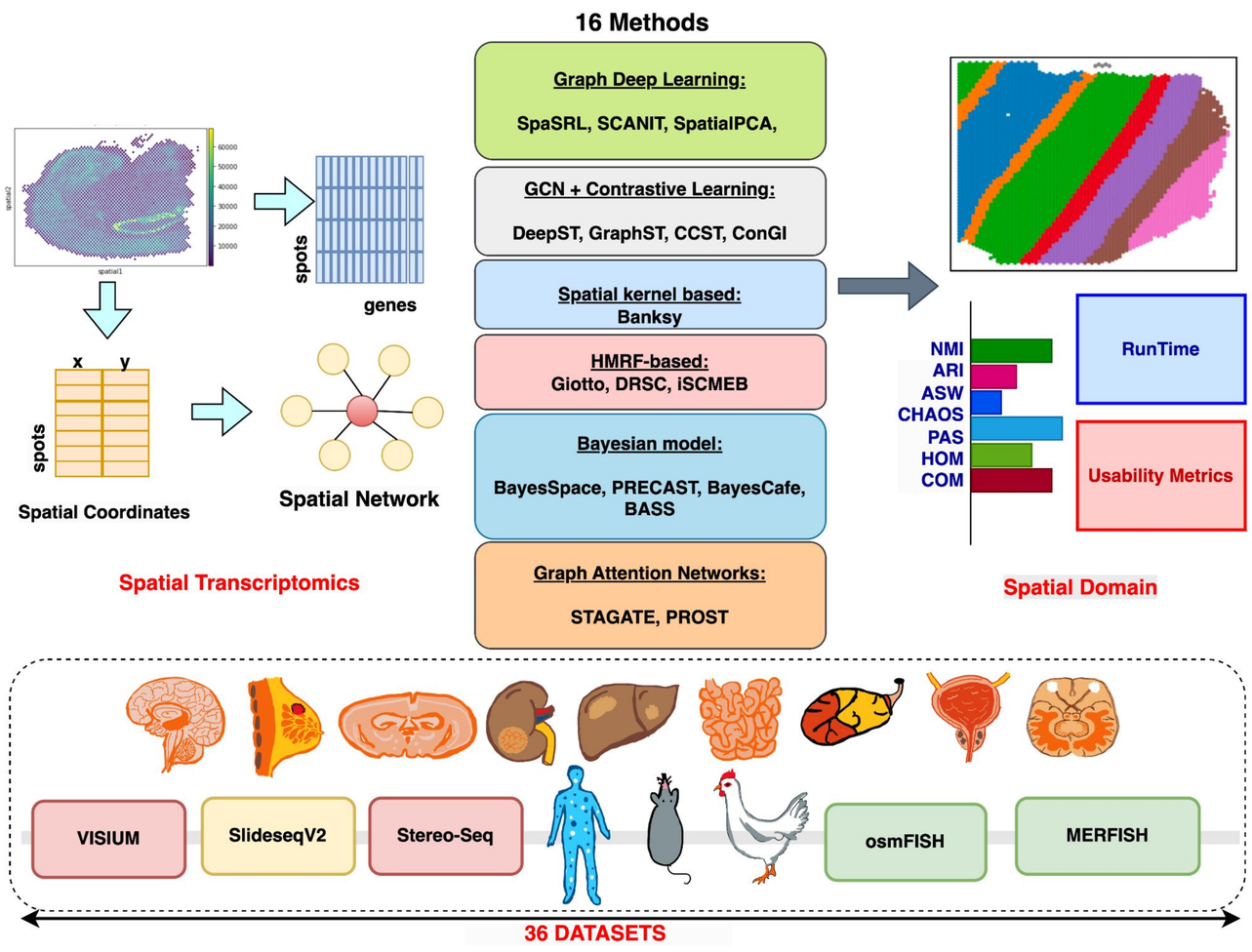
Comprehensive benchmarking of spatial domain detection methods for spatial transcriptomics datasets across technologies and tissue types

Ajita Shree[1], Aditya V[2], Tanush Kumar[2], Hamim Zafar[1, 3, 4, *]

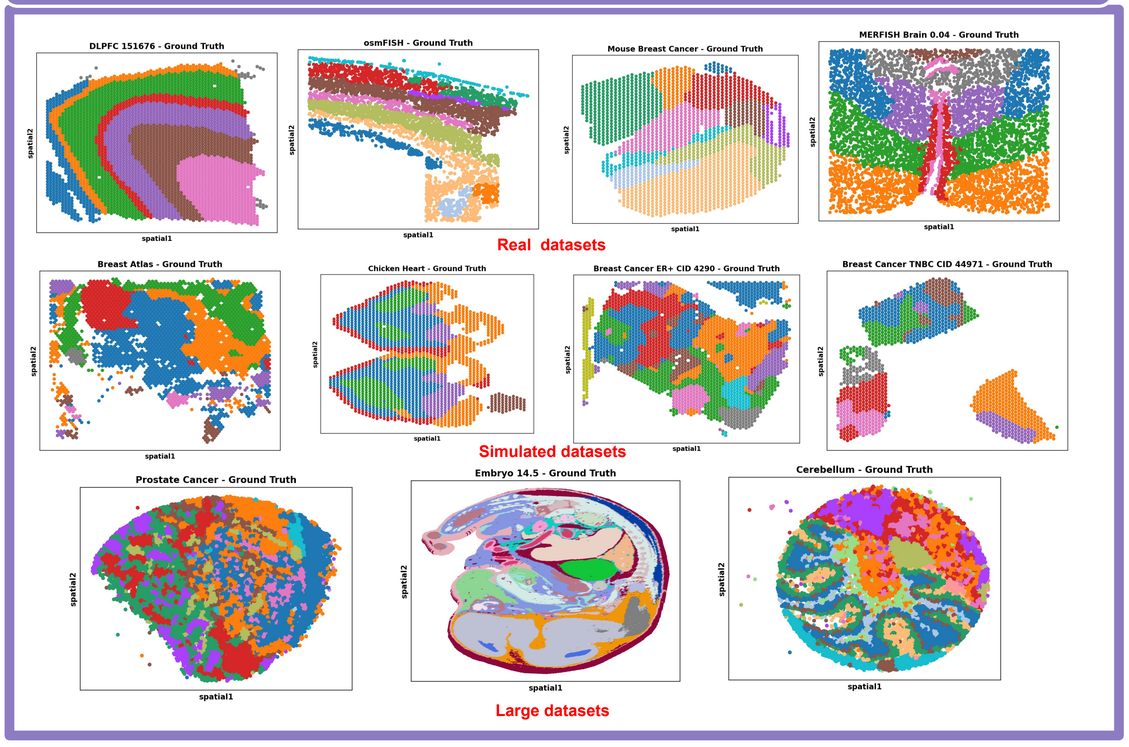
[1]Department of Computer Science and Engineering, [2]Department of Mathematics and Statistics, [3]Department of Biological Sciences and Bioengineering, [4]Mehta Family Centre for Engineering in Medicine, Indian Institute of Technology Kanpur, * Corresponding author, hamim@iitk.ac.in

OVERVIEW

- Rapid development of spatial domain detection methods highlights the pressing need for a comprehensive benchmarking study that not only offers practical guidelines for method selection and application but also informs future methodological advancements in the field
- # Newly added methods : 6; # Different types of tissues 13 as opposed to 7 in previous benchmarking. # Datasets: 36, # Technologies: 5, # Cancer types: 4



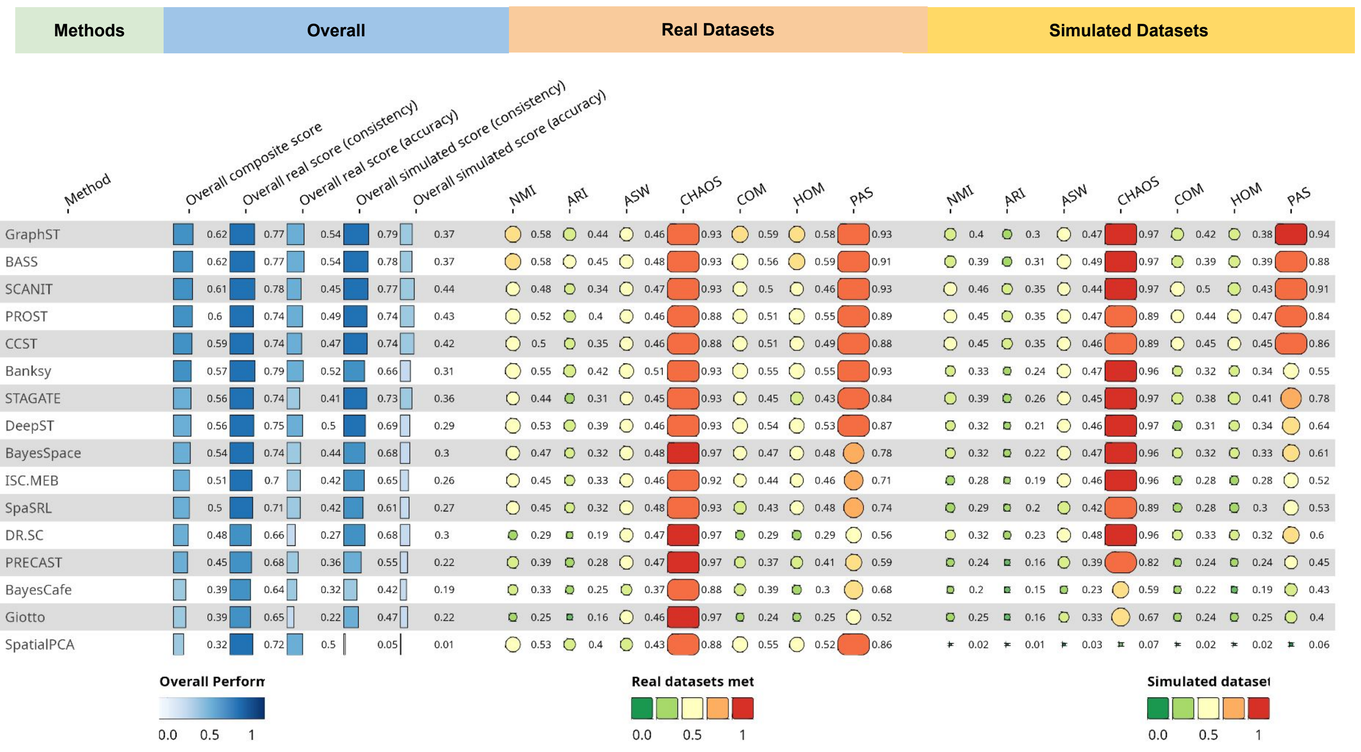
Datasets



Comparison with existing benchmarking studies

Sno	Metric	SD Bench: Nature Methods	Genome Biology	Domain Detection benchmarking
1	**Methods**	11	16	16
2	**Dataset: Tissue types**	7 (Real)	7(Real)	13 => 5 (Real), 8 (Simulated)
3	**Cancer Subtypes**	Yes (2)	Yes (1)	Yes (5)
4	**Large datasets**	1- Embryo	1-SlideseqV2	3 (Embryo, Simulated: Prostate cancer and Cerebellum)
5	**New methods in benchmarking**	NA	6	6 (incl. top performing methods from previous benchmarking)
6	**Details: Real datasets**	34(Brain, Embryo) + 22(7-Breast cancer, 8 Mouse liver, 5 Human liver, 2 Pancreatic cancer)	31(Dlpfc-12, Mouse breast cancer-1, Mouse brain-2, Breast cancer - 8, Hippocampus, Embryo-2, Starmap-1, Merfish-5)	21(Brain, Embryo, Mouse breast cancer)
7	**Details: Simulated datasets**	145(10+10+5+120), Majority Voting algorithm	No	16(Kidney cancer-3, Breast cancer-2, Liver cancer-4, Breast atlas-1, Intestine-2, Chicken heart-1, Prostate cancer, brain)
8	**Datasets: Technologies**	6 (Visium, Stereo-seq, Barista-seq, Merfish, osmFISH, STARMap)	5(Visium, Merfish, Stereo-seq, SlideSeqV2,Starmap)	5(Visium, Merfish, Stereo-seq, osmFish, Slide-seq V2)

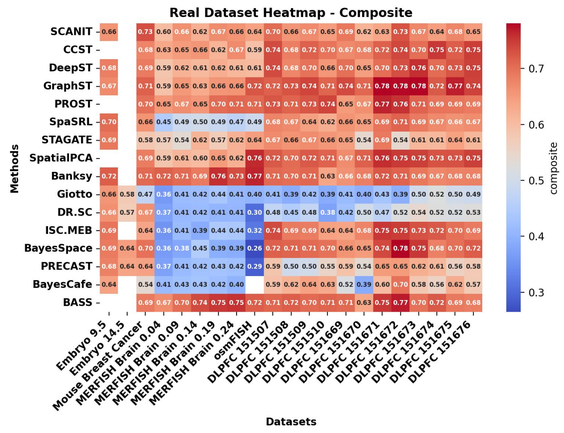
Overall Result



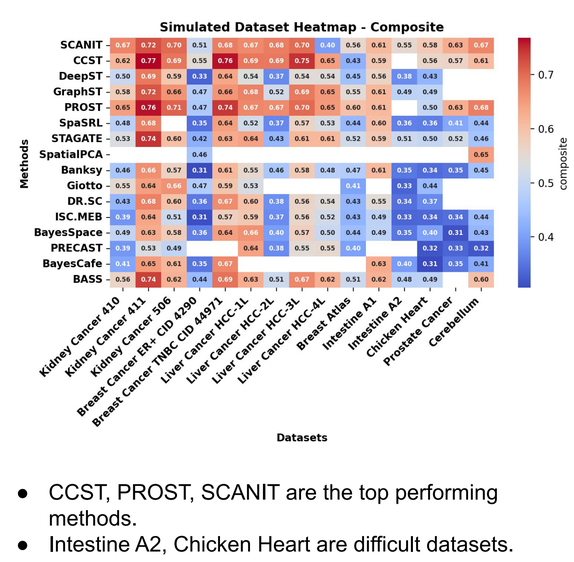
- **GraphST, SCANIT, PROST, BASS, CCST** are the top performing methods
- Newly added methods to benchmarking: spaSRL, PROST, DeepST, Giotto, iSCMEB, BayesCafe

Individual dataset results

Accuracy metrics NMI, ARI, HOM, COM + **Continuity metrics CHAOS, ASW, PAS** → **Overall Composite score**

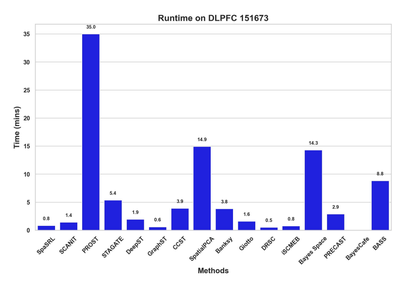


- **DLPFC**: GraphST, SpatialPCA, BASS, PROST, , BayesSpace
- **Embryo**: Banksy, spaSRL, PRECAST
- **osmFISH**: Banksy, SpatialPCA, GraphST
- **MERFISH**: BASS, Banksy, Prost

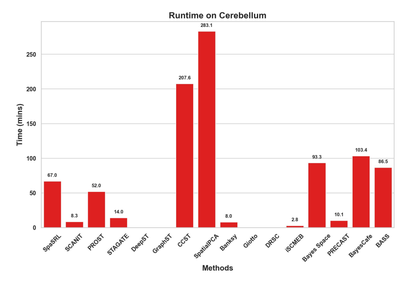


- CCST, PROST, SCANIT are the top performing methods.
- Intestine A2, Chicken Heart are difficult datasets.

Runtime comparison



Fastest methods: DRSC, GraphST, iSCMEB, spaSRL



Fastest methods: iSCMEB, Banksy, SCANIT

Guidelines

Sno	Category	Criteria	1st	2nd	3rd	4th	5th
1	**Tissue**	Brain	BASS	PROST	SpatialPCA	Banksy	CCST
2		Cancer	CCST	PROST	SCANIT	GraphST	STAGATE
3		Organs	SCANIT	GraphST	BASS	STAGATE	DeepST
4		Whole Embryo	BayesSpace	PRECAST	DRSC	Giotto	Banksy
5	**Technology**	Large datasets	BayesSpace	PROST	SCANIT	PRECAST	CCST
6		Visium datasets	GraphST	PROST	CCST	BASS	SCANIT
7		Merfish datasets	BASS	Banksy	PROST	SCANIT	CCST

References

- Yuan et al. Benchmarking spatial clustering methods with spatially resolved transcriptomics data, Nature Methods
- Hu et al., Benchmarking clustering, alignment, and integration methods for spatial transcriptomics, Genome Biology
- Long et al. , Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST, Nature Communications

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